MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY, CHARGE STORAGE

TYPE 7268B

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Direct view charge storage tube, 5-inch, flat face, 2-writing guns, electrostatic focus and deflection, aluminized, P20 phosphor, factory collimated.

DIMENSIONS AND PIN CONNECTIONS: See figure 1.

ABSOLUTE-MAXIMUM RATINGS: Writing guns - each gun (voltages with respect to writing gun cathode)

Parameter:	Ef	Ehk	Ec1 (Note 1)	ec1	Ec3	tk	Ec2/Ec4
Unit:	V	Vdc	Vdc	vdc	Vdc	sec	Vdc
Maximum:	6.93	+125	0 (Note 6)	+2	2000	---	2750
Minimum:	5.67	-125	-200	---	---	30	---
TEST CONDITIONS:	6.3	---	Adjust	---	Adjust	---	2495 to 2505

ABSOLUTE-MAXIMUM RATINGS: Writing guns - each gun (voltages with respect to reference input) (Note 1)

Parameter:	ed	Rg	Zd	Ek	Alt
Unit:	v	Meg	Meg	Vdc	ft
Maximum:	500	1	0.1	---	60,000
Minimum:	---	---	---	-2650	---
TEST CONDITIONS:	(Note 14)	---	0.1	-2400	---

ABSOLUTE-MAXIMUM RATINGS: Viewing gun (voltages with respect to reference input) (Note 1)

Parameter:	Ef	tk	Ebe	Escn	R screen	Ehk	Rbe
Unit:	V	sec	Vdc (Note 2)	Vdc (Note 10)	Meg (Note 3)	V	Ohm (Note 2)
Maximum:	6.93	---	35	11,000	---	+125	5K
Minimum:	5.67	30	-20	---	1	-125	---
TEST CONDITIONS:	6.3	---	2	10,000	1	---	---

ABSOLUTE-MAXIMUM RATINGS: Viewing gun - factory collimated section (voltages with respect to reference input) (Note 1)

	Positive Input Vdc	Negative Input Vdc
Maximum:	300	0
Minimum:	---	-115
TEST CONDITIONS:	(Note 36)	(Note 28)

REQUIREMENT OR TEST: GENERAL

3.2 Qualification - Required (Note 4) --- Aging (Note 30)

E-50.2 Holding period - t = 48 hrs. Section 5 Preparation for delivery

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Qualification inspection					
1006	Salt spray					
----	Operation	(Note 22)				
1136	Rough handling test end points	(Notes 7 and 7a)				
	Quality conformance inspection, part 1 (Note 32)					
5201	Cathode current (writing gun)	(Note 9)	Ik	---	1000	uA
5201	Positive input (viewing gun)		I pos	---	5	mA
5201	Negative input (viewing gun)		I neg	---	1	mA
5201.2	Voltage breakdown					
5201.3	Voltage breakdown (electrostatic types)	(Note 10)				
----	Storage time	(Note 11)		15	---	sec
5101	Alinement, base	3D4, base key				
5106	Blemishes	(Note 12), figure 2				
----	Loose particles	(Note 34)				
5221	Light output	(Note 31)		2200	3700	ftL
----	Resolution	(Note 15)		70	---	lines per inch
5241	Grid cutoff voltage (writing gun)		Ec1	-100	-60	v
5246.1	Focusing voltage at cutoff			250	425	v
5251.2	Grid No. 1 leakage			---	1	uAdc
5251.5	Focusing electrode leakage (grid No. 3 of writing guns)			---	5	uAdc
----	Insulation of electrodes	(Note 20)		100	---	Meg

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Quality conformance inspection, part 1 (Note 32) (Continued)					
----	Screen current	Saturated light output		---	700	uA
----	Heater cathode warm up time	Stable operation (Note 35)		---	60	sec
-----	Grid drive requirements (writing gun B)	(Note 21), figures 3 & 6		20	---	%
----	Writing uniformity (writing gun B)	(Note 21), figures 3 & 6		---	2.0	ratio
----	Useful scan			3.5	---	inches
----	Vibration	(Note 25), figure 4		---	1	mm

METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP. LEVEL OR CODE	SYMBOL	LIMITS MIN.	LIMITS MAX	UNIT
	Quality conformance inspection, part 2							
Appendix D, 20(b)	Dimensions	See figure 1	6.5	L6				
----	Interaction factor	(Note 23)	6.5	L6			$1x10^{-5}$	in/ Vdc
1101	Base, cap, and insert secureness	(Note 7) immersion not required	6.5	L6				
----	Grid drive requirements (writing gun A)		1.0	II				
----	Writing uniformity (writing gun A)		1.0	II				
1141	Pressure	(Note 7)	6.5	L6				
1301	Heater current (each gun)		1.0	II	If	0.5	0.7	A
----	Magnetic shielding	(Note 26), figure 1	6.5	L6				
5101.7	Angle between traces		1.0	II		89	91	degrees
----	Half tones		1.0	II		3	---	

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP. LEVEL OR CODE	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Quality conformance inspection, part 2 (Continued)							
5101	Trace alinement between corresponding traces		1.0	II				
5251.6	Spot displacement	(Note 16)	1.0	II		---	1	mm
----	Collimation assurance	(Note 5)	6.5	L6		---	---	
----	Erase time	(Note 29)	6.5	L6		15	45	ms
5231.2	Spot position		1.0	II		---	15	mm
5248.1	Deflection factor	1D2	1.00	II		82	100	V/m 2.5KV
5248.1	Deflection factor	3D4	1.00	II		82	100	V/m 2.5KV
5248.2	Deflection factor uniformity		1.0	II		---	7	%
----	Pattern distortion	(Note 19)	1.0	II				
----	Useful viewing diameter	Concentric with shield	1.0	II		4	---	inches
----	Erasing	(Note 13)	1.0	II				

METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Quality conformance inspection, part 3 (Life tests)					
4.7	Life test	Note 7) light output= 2200 ftL; t= 500 hrs (min) Group C				
4.7.3	Life test end points	Light output		2000	---	ftL
		Cathode current (Note 9)				

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Quality conformance inspection, part 3 (Note 8) (Periodic check tests)					
1331	Direct inter-electrode capacitance	Cathode to all Writing gun	C	---	8	pf
		Grid 1 to all Writing gun	C	---	15	pf
		D1 to D2	C	---	3	pf
		D3 to D4	C	---	2	pf
		D1 to all	C	---	10	pf
		D2 to all	C	---	10	pf
		D3 to all	C	---	10	pf
		D4 to all	C	---	10	pf
----	Low temperature operation	(Note 33)	T	-65	---	°C
----	Low pressure-High temperature	(Note 8A)				
----	Vibration	(Note 17)		---	1	mm
----	Post vibration limits	(Note 17)				
----	Temperature cycling	(Notes 7 & 24)		5	---	cycles
5201	Focusing electrode current (each gun)		I_c	-10	+5	μA
5246.2	Focusing voltage at modulation condition	(Note 18)		75	425	V
5251.1	Heater cathode leakage (writing gun)	Ehk = ± 100 Vdc		---	10	μAdc
5201	Heater reference input (viewing gun)		I_{ref}	---	5	mA

NOTES:

1. The factory collimated section includes all viewing gun electrodes, except screen, backing electrode, and heater. G2 and G4 of both writing guns may also be included in this section. Care shall be taken, however, to insure that the G2G4 (W1W2) combination is maintained at 100 ± 5 volts with respect to reference input potential which shall be grounded.
2. Backing electrode resistance is Rbe; backing electrode potential is Ebe.
3. Series current limiting resistance in screen circuit.

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NOTES - Continued:

4. The activity responsible for the qualified products list is the Naval Ship Engineering Center, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification." (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pa. 19120).

5. The tube shall be operated at the conditions specified for the Writing uniformity test in Note 21. The voltages applied to the tube, shall be varied, one at a time, as required by the following steps. At the voltage specified in each step, the writing uniformity of the tube shall be evaluated by the procedure of Note 21. The writing uniformity at each step shall be within the limit specified in the writing uniformity test. At the end of each step, the voltage which was changed shall be returned to the condition specified for Note 21. Erase pulse amplitude and noise input may be adjusted for each step to obtain the erase time and meter M1 reading required for the writing uniformity test, but no other voltages shall be changed.

   Step 1: Raise view heater voltage to 6.93 volts.
   Step 2: Reduce view heater voltage to 5.67 volts.
   Step 3: Replace the circuit of Note 28 with a low impedance supply of -50 volts at the negative input.
   Step 4: Replace the circuit of Note 36 with a low impedance supply of -70 volts at the negative input.
   Step 5: Replace the circuit of Note 36 with a low impedance supply of 235 volts at the positive input.
   Step 6: Replace the circuit of Note 36 with a low impedance supply of 270 volts at the positive input.

6. The writing-gun grid No. 1 shall never be more positive than necessary to write the display to saturated brightness for a given scanning and drive condition.

7. Operable rejects from the current lot may be used for appropriate Quality conformance inspection, part 2 tests and Rough-handling test end points. Rejects for mechanical defects only may be used for life tests.

7a. After the Rough-handling test end points, the tube shall meet the following requirements:

   1. There shall be no damage to external parts of the tube.
   2. Tubes shall meet all electrical parameters of this specification.
   3. Tubes shall be vibrated in accordance with Note 25.

8. Periodic environmental tests may be conducted on tubes which are rejected in the normal course of production testing, provided the item causing rejection is not of a nature to compromise the results of environmental testing and provided the defect is recognized in evaluating performance during or after the test sequence. From each group of 50 tubes produced (or from each monthly production lot, if the lot exceeds 50 tubes) select one tube to be evaluated against these tests. One tube may be subjected in sequence to all tests or a separate tube may be selected for each test if concurrent testing is convenient. If this initial sample passes each required test, the group of 50 tubes (or the monthly lot) may be shipped. If a test is failed, two additional samples must be subjected to that test and pass the test to permit release of the group which they represent. If either or both of these two samples fail, 100 percent product evaluation is required until such time as ten successive tubes pass. (One hundred percent testing for the vibration test is not allowed.)

8a. The tube shall be conditioned for a period of not less than one hour in a chamber having an ambient air temperature of +100°C. At the end of one hour, the tube shall be operated with normal voltages applied and at a pressure equivalent of 32,000 feet. Following this the chamber temperature shall be reduced to +53°C. At the end of one hour, the tube shall be operated with normal voltages applied and a pressure equivalent of 60,000 feet altitude. Connections shall be made to the base pins in a manner that does not degrade the tube breakdown characteristics. The criteria of satisfactory operation for a period of 15 minutes shall be the presence of a clearly defined circular trace display (absence of corona and arc-over).

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NOTES - Continued:

9. Writing from visual extinction with no erase pulse applied, write a single frame of pattern with writing speed of 36,000 inches per second, with pattern size adjusted so that lines are just merged (see Method 5221). With Ecl voltage (not to exceed 50 volts above cutoff) adjust for brightness of at least 2000 ftL, measured within five seconds after cessation of writing, the cathode current shall not exceed 1000 uAdc.

10. With the writing guns cutoff under test conditions, operate viewing gun under the following conditions for a minimum of five minutes:

Ef: 6.3 volts	BE: 2Vdc
Negative input: -115 Vdc	E scn: 11,000 Vdc
Positive input: 300 Vdc	

At the completion of the specified time, Escn shall be increased to 14,000 Vdc for 60 seconds. There shall be no recurrent breakdown between elements.

11. Write the viewing screen to saturation; increase the erase pulse amplitude to just effect visual cut-off; remove the erase pulse; no area shall reach 10 percent of saturation within 15 seconds as measured with a Spectra Brightness Spot Meter (Photo Research Corporation, or equivalent) at a distance of 5 feet (0.5 inch viewing diameter). Incident light on the area under test shall be less than 1 percent of saturated light output.

12. (a) Dark spots. - Dark spots are defined as areas of the screen which do not brighten during writing, or which require unusual processes to effect brightening; areas that lack persistence or whose persistence is obviously less than that of surrounding areas to an observer, and includes the more easily recognized dead spots, burned spots and areas damaged by arc-overs. The maximum number of dark spots shall be as follows:

Maximum diameter of spot (inch)	Number of spots in 1/		Total in Zones 1, 2, and 3
	Zone 1	Zones 2 and 3	
0.015 - 0.030	2	2	4
0.030 - 0.040	2	2	2

1/ Dark spots are permissible in zones 4, 5, 6 and 7 but shall not influence zones 1, 2 or 3. Dark spots less than 0.015 inch are permissible provided their minimum separation is 0.3 inch or the total number in zones 1, 2, and 3 combined does not exceed 10.

(b) Light spots. - The two types of light spots are as follows:

(1) Bright spots. - Small areas of the screen which are brighter than the background, but not as bright as maximum brightness of the surrounding screen area. Operating within the limits of an erase-pulse generator whose output is a rectangular 60 ± 10 us wide, positive going pulse, adjustable between 0 and 12 volts peak-to-peak, 300 pulses per second, this type of spot will be visible when the erase pulse amplitude is that required to just erase the background (exclusive of the light spot) to cutoff, will be eliminated by increasing the amplitude an additional 10 percent.

(2) Brilliant spots. - Small areas of the screen which are as bright or brighter than maximum brightness of the surrounding screen area. This type of spot cannot be eliminated by the 10 percent increase specified in (b) (1), and includes all light spots not specified in (b) (1). Light spots shall be observed during process of erasing from saturated light output to background level and shall be measured when the display reaches background level.

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	Maximum diameter of spot (inch)	Number of spots in 1/ Zone 1	Zones 2 and 3
Brilliant	Any size	None	None
Bright	Greater than 0.030	None	None
Bright	0.030 or less	1	3 total

1/ With a total of 4 in zones 1, 2 and 3. Bright spots in zones 2 and 3 shall not influence zone 1.

Light spots are permissible in zones 4, 5, 6 and 7 but shall not influence zones 1, 2 or 3.

13. Erasure shall be accomplished by applying rectangular pulses to the backing electrode. Incident light on the area under test shall be less than 1 percent of saturated light output. Write the useful screen area to equilibrium brightness. Apply erase pulses of width 60 ± 10 us and prr of 300 pps. Erase time of 3.5 ± 0.5 seconds shall be obtained when the erase pulse amplitude is varied in the range 2 to 10 volts peak-to-peak. The difference between the amplitude of an erase pulse required to brighten any area of the unwritten screen and the amplitude of an erase pulse required to completely illuminate the screen shall not exceed 20 percent of cutoff.

14. The average voltage of each pair of deflection plates shall be 100 ± 5 Vdc with respect to the reference input. The potential of G2G4 (W1W2) shall also be 100 ± 5 Vdc with respect to reference input.

15. To measure direct view storage tube (DVST) resolution in terms of lines per inch at the same time that the light output brightness is equal to 1000 ftL, the procedure is as follows:

Adjust erase time for 3.5 ± 0.5 seconds. Apply a 45 to 80 line raster pattern to the tube by connecting a 60 cps sawtooth voltage to the vertical axis and 2700 to 4800 cps sawtooth voltage to the horizontal axis reflection plates of one of the writing guns. (Note: The selection of a specific number of lines will be determined by the size of the area measured by the type of lightmeter being used.) Set the raster trace length at 3 1/2 inches. Expand the low frequency scanning amplitude and focus at the center of the raster. Compress the raster until the lines just merge and the line structure disappears. A photometer, corrected for the spectral response of the eye and having the diameter of the aperture of its light sensitive area smaller than the narrower dimension of the collapsed raster, shall be used to measure the light output. Set the light level at 1000 ± 5 percent ftL at the center of the raster.

(Note: It may be necessary to further collapse or expand the raster by changing the amplitude of low frequency scan while adjusting light output in order to obtain a properly merged raster at the desired brightness level.)

Again expand the pattern and re-adjust focus at the center of the raster if necessary. Compress the raster until the lines just merge at the center of the pattern. At the proper degree of merging, the distinguishable line structure just disappears and bright lines do not appear. If the raster is incorrectly overmerged, there will result a reverse line structure consisting of brighter lines where the dark lines were prior to merging. Measure the width of the compressed pattern at the center of the raster. The resolution is then given by the quotient of the number of lines in the raster divided by the measured width of the compressed pattern in inches. Interchange the connection of deflection elements to the low and high-frequency scanning supplies and re-measured resolution without re-adjustment of focus. The above procedure, performed at points on the center vertical axis, 1 1/2 inches above or 1 1/2 inches below the center of the tube face, shall yield resolution measurements greater than or equal to the minimum limit specified herein. Without refocusing, repeat for alternate axis. Repeat all of the above procedure for the other writing gun.

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16. Adjust erasure to a convenient value. Connect each deflecting electrode through a separate 100K ohm resistor to G4 of the writing gun. With means for short circuiting each resistor, measure the spot displacement when each resistor is short circuited in turn.

17. The tube shall be suitably mounted on the vibration table by large base diameter and zone C (see figure 1) and tested as follows:

(a) With the tube mounted and with a circular trace positioned on the tube face and operating at prevailing room temperature, the table shall be vibrated along the X-X axis with the frequency varying between 10 and 500 cps at an amplitude as indicated by vibration specification curve (see figure 4). The rate of frequency change shall be approximately uniform, and shall be such that a complete cycle, for example, 10 to 500 cps, will consume approximately 15 minutes. The test shall continue for 90 minutes. During this time all resonant points shall be noted and the increase in linewidth of the circular trace shall be within the limits specified herein to accurately determine resonant points, the frequency shall be varied slowly through the range around each of the noted resonant points. A resonant point is defined as that frequency at which a noticeable change in the written or scored information on the screen in visible. Tubes shall be rejected for occurrence of permanent brilliant or dark spots exceeding blemish limits, or for other indications of tube malfunction or degradation in performance.

(b) With the tube operating, the table shall be vibrated for 30 minutes at the resonant frequency noted in step (a). Where more than one resonant frequency is noted, the test period may be divided among the resonant frequencies, whichever is considered most likely to produce failure. Where resonant frequencies are not apparent within the specified frequency range, the tube shall be vibrated for a period of one hour at a frequency of 55 cps. During this time, the increase in linewidth of the circular trace shall be within the limits specified herein. Tubes shall be rejected for occurrence of permanent brilliant or dark spots exceeding blemish limits or for indications of tube malfunction or degradation in performance.

(c) Steps (a) and (b) shall be repeated with the tube mounted such that the vibration is along the Y-Y axis.

(d) Steps (a) and (b) shall be repeated with the tube mounted such that the vibration is along the Z-Z axis.

(e) Following vibration, retest tube. All requirements and test limits specified in Quality conformance inspection, part 1 (except vibration) and the 1 percent AQL tests in Quality conformance inspection, part 2 of this specification shall be satisfied. In the event a reject tube has been used for this test, under the condition outlined in Note 8, the item which was cause for initial rejection may be disregarded. (Note: This test is classified as a destructive test, in the sense that tubes subjected to it may not be shipped as good product.)

18. Each gun: From cuttoff to +50 volts above cutoff: $\Delta Ec3 = 200$ Vdc (max). Spot shall be pulsed to prevent damage to the tube.

19. The total movement of the ends of a 2.8 inch trace (centered with respect to the tube face), deflected horizontally or vertically 1.40 inches each side of the center of the tube face, shall not exceed 0.060 inch. The 1D2 trace shall be considered horizontal and the 3D4 trace shall be considered vertical.

Alternate method for measuring patter distortion. - A rectangular raster shall not fall outside a square frame whose inside dimension is 2.74 inches and outside dimension is 2.86 inches and whose center coincides with the centerline of the tube.

20. The dc resistance between any element, exclusive of heaters and elements included within the factory collimated section but including the magnetic shield, and all other elements connected together measured at a minimum potential of 250 Vdc shall be no less than specified herein. Should the polarity of the measuring potential influence the resistive reading, the lower of the two readings shall apply. The heaters need not be energized during this test.

21. During the Grid drive requirements (writing gun B) test, performance is judged by feeding a random noise signal of known amplitude to each of the two writing guns in succession and measuring the resulting display luminescence as a percentage of saturated light output. The Writing uniformity

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NOTES - Continued:

21. (Continued)
(writing gun B) test shall be performed immediately following and in conjunction with grid drive. The equipment to be used, the conditions of test, the calibration procedure, and the test sequence shall be as follows:

(a) Equipment:

(1) Random Noise Generator (5 Mc scale) - General Radio type 1390-B, or equivalent.

(2) Sine Wave Generator (5000 to 9000 cps) - Hewlett-Packard Model 200 ABR or equivalent. (Note: Sine wave shall be non-synchronous with vertical pulse repetition frequency (PRF)).

(3) Video gating and mixing circuitry designed to accept, successively, the output from the random noise and sine wave generators. These signals shall be gated on only during the mutual period of the vertical scan and the pedestal. Negative-going portions of noise shall be rejected; for example, clipped, and working into a 2K + 5 percent load. Noise and pedestal should add. Do not clip calibrating sine wave. Response of the system up to and including the DVST writing gun grid shall be 2 Mc minimum at the 3 db point.

(4) Appropriate scanning waveform, pedestal, and erase pulse generators. (See 21 (b)).

(5) Noise calibrating circuitry in accordance with figure 3.

(6) Light output measuring device in accordance with figure 3.

(b) Conditions of test

(1) Vertical scan height: 3.5 inches (centered).
(2) Vertical scan time: 625 us (linear sawtooth).
(3) Vertical PRF: 500 pps.
(4) Horizontal scan width: 3.5 inches (centered).
(5) Horizontal scan rate: 3.5 in/sec. (linear triangular).
(6) Mean deflecting electrode potential: 100 ± 3 percent Vdc.
(7) Writing gun G1 input: Output waveform from noise calibrating circuitry clamped to G1 bias level.
(8) Noise calibrating circuitry video inputs to point A:
a. Pedestal only (625 us rectangular pulse coincident with vertical scan period), see figure 6, view A.
b. Pedestal plus gated sine wave, see figure 6, view B, or
c. Pedestal plus gated and clipped noise, see figure 6, view C.
(9) Noise calibrating circuitry plus input to point B: Pedestal only.
(10) Erase pulse: Positive going rectangular pulse, 60 ± 10 us in width, at a PRF of 300 pps.
(11) Other tube voltages in accordance with test conditions.

(c) Calibration procedure

(1) With pedestal only applied to point A, adjust pedestal amplitude for 10 volt peak-to-peak reading at point Eo of Figure 3. No further adjustment is permissible.

(2) With 12-volt peak-to-peak pedestal at input B, adjust R7 of noise calibrating circuitry for null reading on meter M1. No further adjustment permissible.

(3) Apply sine wave to pedestal; for example, pedestal plus gated sine wave shall be applied to point A. (See figure 6, view B.) Adjust amplitude of gated sine wave to 23 ± 0.5 volts peak-to-peak as indicated by a calibrated scope at point Eo of figure 3. Read and record meter M1. Remove gated sine wave from pedestal.

(4) Cover opening to photo electric cell V3, close switch S1, and adjust potentiometer R15 for null reading of meter M2.

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NOTES— Continued:

21. (c) (Continued)

(5) Write display to saturated brightness by reducing erase pulse amplitude to zero, adjusting writing bias so that pedestal writes fully a "stored" blank raster, and then adjusting write bias to cutoff.

(6) Place photo electric cell V3 against the faceplate of the DVST and after steady reading is obtained, read and record meter M2.

Note: Avoid saturation of the measuring device by the addition of sufficient neutral density filters in front of the photo electric cell. A reading of approximately 4.5 volts indicates saturation. Sufficient filtering to obtain a reading between 3 and 4 volts will be required.

(d) Test sequence

(1) Apply erase pulse as in (b) (10). Amplitude should be between 2 and 10 volts, peak-to-peak.

(2) Adjust bias of writing gun G1 under test, with pedestal only, until meter M2 reads the same as step (c) (6).

(3) Turn off writing gun when sweep is in center and note the time required for saturated light output to be erased to background brightness (complete erasure over entire face of tube.)

(4) Repeat steps (d) (2) and (d) (3) adjusting erasure pulse amplitude to obtain 3.5 $\pm$ 0.5 seconds erase time.

(5) Apply pedestal only to point A. Adjust bias of writing gun G1 under test just to visual cutoff.

(6) Apply pedestal plus gated and clipped noise to point A. (See figure 6, view C.) Adjust amplitude of noise until meter M1 reads the same as in step (c) (3). If noise peaks exceed 32V, CK1, CK2, CK3 or CK4 of figure 3 may be defective.

(7) Place photo electric cell against faceplate of DVST. Close S1 momentarily. Release S1 and allow sufficient time for integration of noise to reach equilibrium. Scan face of the DVST with photocell to find and record the minimum and maximum readings of M2.

(8) If the minimum reading observed in step (d) (7) is less than 50 percent of meter reading in step (c) (6), the ratio of maximum to minimum reading in step (d) (7) shall not exceed the value specified herein for writing uniformity.

(9) With the photocell located at the minimum position determined in step (d) (7), adjust the calibrated noise input to obtain a value which is 40 percent of the meter reading in step (c) (6). Again scan the face of the DVST for minimum and maximum values. The ratio of maximum to minimum readings shall not exceed the value specified herein for writing uniformity.

(10) If the minimum reading in step (d) (7) is greater than 50 percent of the meter reading in step (c) (6), reduce the calibrated noise input to obtain a value which is 40 percent of the meter reading in step (c) (6). Again scan the face of the DVST for minimum and maximum values. The ratio of maximum to minimum readings shall not exceed the value specified herein for writing uniformity.

(11) Remove noise input and note time for noise or build up to be erased to background brightness (complete erasure over the entire face of the tube). The time shall be not greater than 5 seconds. After this check, close S1 momentarily and release. Place photocell against faceplate of DVST. Allow sufficient time for meter M2 to reach equilibrium. Read and record M2. This reading shall be less than 5 percent of reading of (c) (6).

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NOTES - Continued:

22. As part of qualification, two tubes shall be operated in an APQ-72, in an APQ-100, and in an APQ-109 system for a period of six hours in each, and shall demonstrate satisfactory performance in the systems.

23. The deflection per unit volt of one beam at any position of the beam within its useful scan when balanced voltage of at least ± 300 Vdc applied to the 1D2 deflection electrodes of the other electron gun shall be not greater than the value specified herein.

24. All heaters shall be energized with nominal heater voltage during that portion of each cycle beginning at low temperature equilibrium and ending at high temperature equilibrium. Use MIL-STD-202, Method 107, condition B, except temperature in step 3 shall be 100°C ± 3°C.

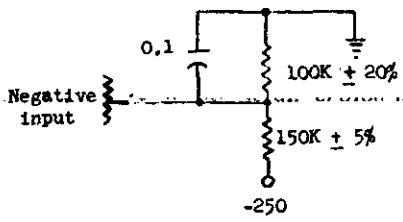
25. This is a 100 percent test and shall be applied to all tubes. With the tube mounted as specified in Note 17, with the vertical reference trace positioned vertically, and operating at prevailing room temperature, the table shall be vibrated along the Y-Y axis with the frequency varying between 10 and 500 cps at an amplitude as indicated by the vibration specification curve (see figure 4). The rate of frequency change shall be approximately uniform, and shall be such that a complete cycle (10 to 500 to 10 cps) will consume approximately 10 minutes. The tube shall be vibrated for one cycle. During this period, maximum particle activity shall be observed and recorded in the lower and upper frequency ranges (10 to 100 cps and 100 to 500 cps). Select a single fixed frequency in each of the two ranges where maximum particle activity was observed and vibrate the DVST for an additional five minutes at each of the frequencies. Tubes which display blemishes on the screen, show arcs or flashing, or develop permanent or temporary shorts, shall be rejected. Following vibration, retest the tube. All initial requirements and test limits shall apply and the tube shall be within initial limits.

26. The shielding shall be such that a 10 gauss field will not deflect an undeflected spot more than 1 mm and will not otherwise affect the performance of the storage tube. The magnetic shield shall be secured to the tube at least to the full length of zones A, B and C, as shown on figure 1 with an encapsulating material, electrically non-conducting and non-nutrient. The shield shall be capable of withstanding a magnetic field when tested by the following or equivalent procedure. A suitable pair of air-core coils to produce the magnetic field for testing has 375 ampere-turns each, a mean cross-section of 10.5 by 2.75 inches perpendicular to the field and a separation of 8.5 inches. The center of the coils shall be approximately centrally located about the tube.

With the tube operating as for the spot position test, measure spot motion when the coil current is interrupted. The spot motion under these conditions shall not exceed 1.0 mm. Note any effects of the magnetic field on the viewing beam.

27. At the end of 500 hours, the tube shall be retested and shall meet initial requirements of Quality conformance inspection, part 1, except light output and vibration.

28. With circuit shown below:



NOTE:
No damage should result to tube when control is moved over complete range of adjustment. Test conditions for negative input obtained when the wiper is at the junction of the 150K resistor and the potentiometer.

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NOTES - Continued:

29. With display at equilibrium (as defined in Note 31) apply a series of rectangular pulses to the backing electrode at a PRF of about 2 per second and at a pulse width of about 5 ms. Note how many pulses it takes to just completely erase center of display. Multiply the number of pulses to completely erase center of display by the pulse width to obtain erase time in milliseconds. Increase or decrease amplitude of rectangular pulse slightly and repeat above until pulse amplitude is obtained which gives lowest value of erase time. The erase time measured at this pulse amplitude shall be within the limit specified herein.

30. Prior to the performance of any tests, the tube shall be operated for a minimim of 48 hours.

31. Messured at equilibrium with Spectral Spot Brightness meter or equivalent. Light output is measured over any 1/2 inch or greater area any place within the useful viewing diameter. Equilibrium is that level of brightness obtained several seconds after cessation of writing with no erase pulse applied and where the entire screen area has been written to the maximum obtainable brightness (saturation brightness).

32. Data on all Quality conformance inspection, part 1 tests for every tube shall be submitted to the Government inspector with the tubes.

33. Tubes shall be cold soaked for a period of 48 hours at -65°C. At the end of this period and while still at the low temeprature, all heater voltages shall be applied for a period of 15 minutes. Tubes shall then be removed from freezer and tested for storage time.

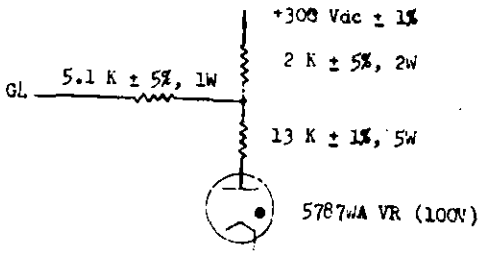
34. While no clearly defined rules can be offered for loose particle content in the DVST, the following guide lines shall apply:

    (a) Tubes which exhibit momentary bursts of particle activity on the application of voltage and in which particles show no clinging action to cause blemishes, shall be considered acceptable provided the tubes pass vibration test, Note 25.

    (b) Tubes which exhibit particle activity on the application of voltages or under vibration to a degree which causes blemishes, arcing, repetitive flashing, or other disturbance not normal to storage tube operation, shall not be considered acceptable.

35. After 60 seconds of heater-cathode warm-up time, application of normal tube voltage will produce a collimated 4-inch (min) display.

36. With series current limiting resistance of 5100 ohms, in circuit shown below:



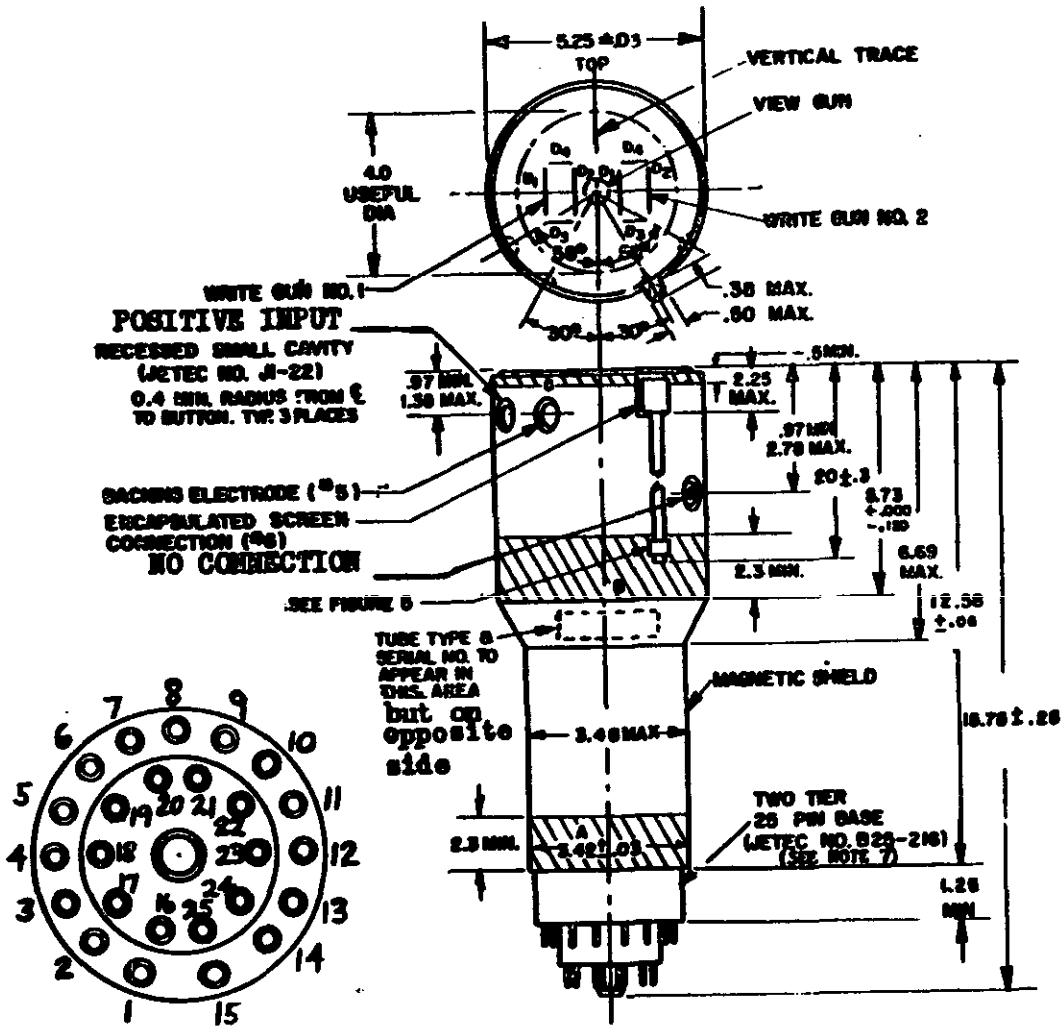
37. CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Review activity:
Navy - SH

User activities:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - SH
(Project 5960-N722(NAVY))

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BASE PIN CONNECTIONS

PIN NO.	ELEMENT	PIN. NO.	ELEMENT
1	D2(W2)	16	H(W2)
2	D1(W2)	17	G1(W2)
3	D3(W2)	18	H(W2)
4	D4(W2)	19	G3(W2)
5	MAG. SHIELD	20	K(W2)
6	NEG. INPUT	21	H(W1)
7	REF. INPUT	22	G1(W1)
8	H(V)	23	H(W1)
9	H(V)	24	G3(W1)
10	SEE NOTE 10	25	K(W1)
11	D4(W1)		
12	D3(W1)		
13	D2(W1)		
14	D1(W1)		
15	NC		

(W1) - Write gun No. 1
(W2) - Write gun No. 2
(V) - View gun

Figure 1 - Outline drawing for tube type 7268B.

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NOTES:

1. Centerline of base axis shall be not more than 0.062 T. I. R. from a perpendicular erected at center faceplate.
2. Faceplate shall be even with shield or extend beyond shield by less than 0.063.
3. Angular tolerance shall be ± 3 degrees.
4. Shield finish: Prime as required and finish type III or IV (black) in accordance with MIL-E-5557.
5. Cylindrical sections at large diameter of base and zones B and C shall be concentric with zone A within 0.062 T. I. R. of surface gage.
6. Volume between glass and shield in zones A, B, C around electrodes 3, 4, 5, and base shall be potted except that area around screen connection need not be potted.
7. Base insulating material shall be diallyl phtalate, in accordance with type SDI-5 of MIL-M-14.
8. Weight of tube shall be 5 1/4 ± 1/4 lb.
9. Screen backing electrode, positive input, and no connection terminal shall be identified on the side of the shield as SCRN (#6), BE (#5), CLT (#4), and CLM (#3), respectively.
10. Alternate connection: NC or G2G4 (W1W2).

Figure 1 - Outline drawing for tube type 7268B. (Contd.)

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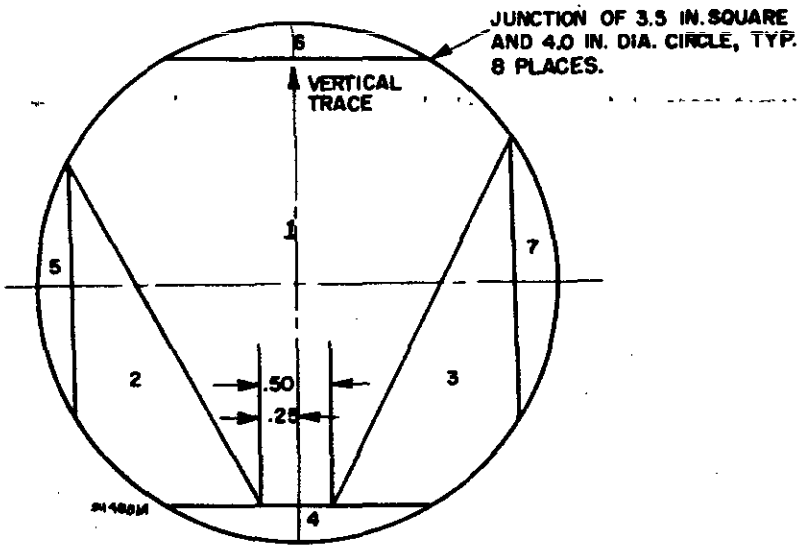


Figure 2 - Blemish zones.

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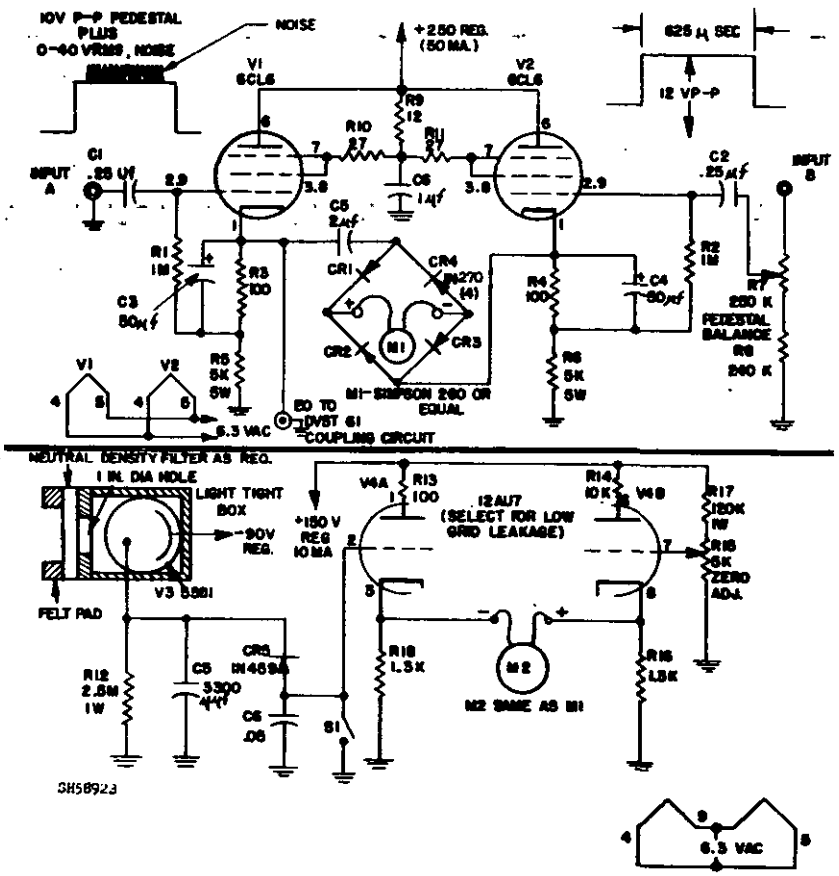


Figure 3 - Grid drive test equipment.

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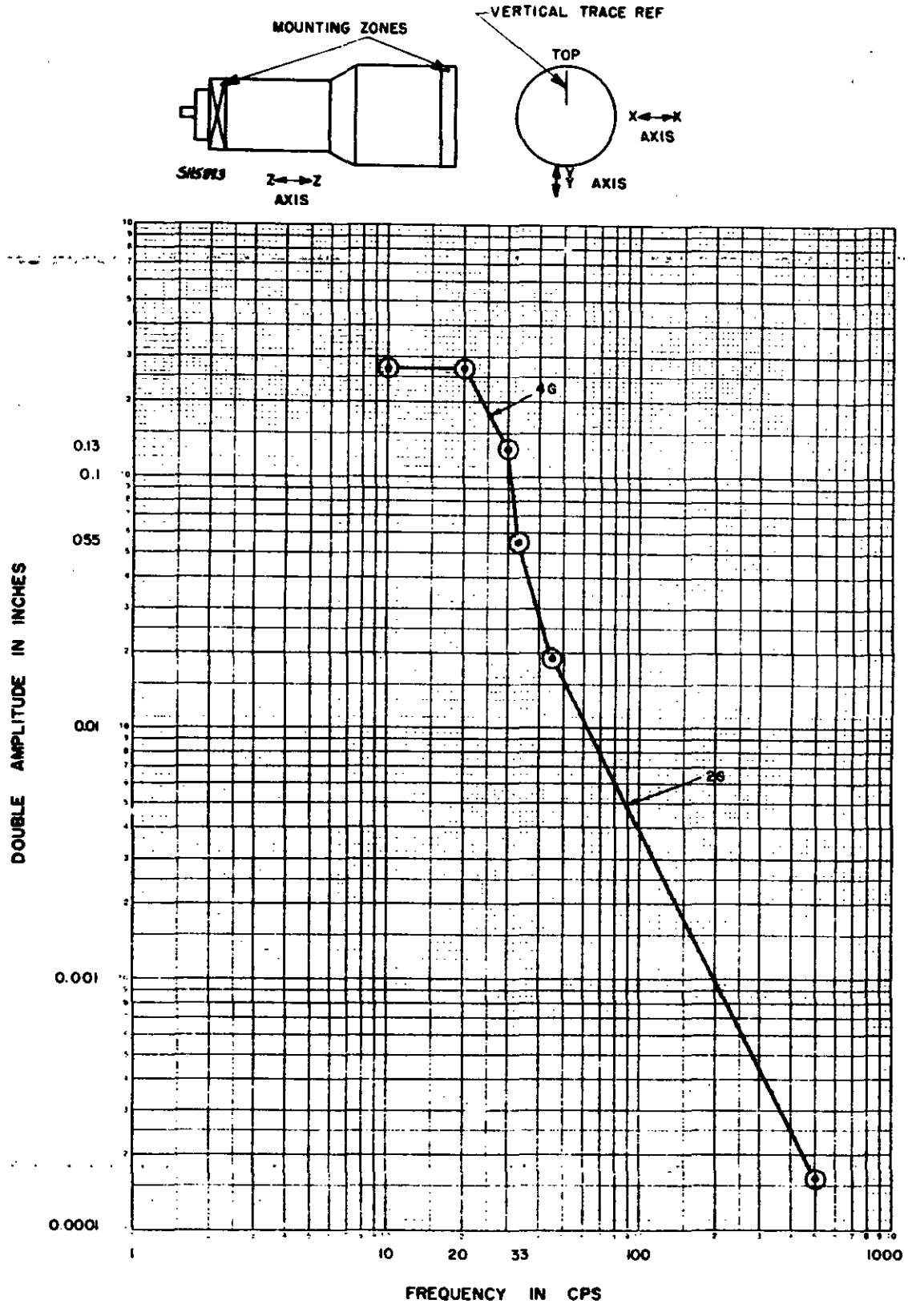


Figure 4 - Vibration requirements for tube type 7268B.

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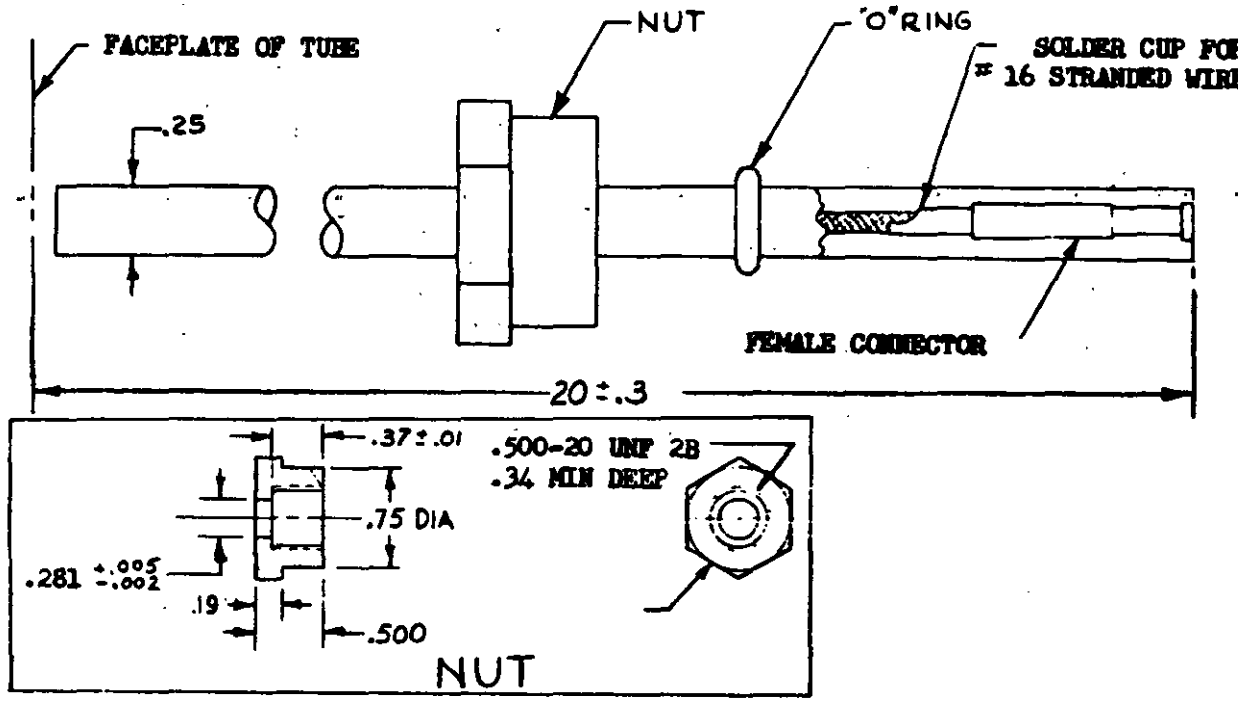


Figure 5 - Screen connection for tube type 7268B.

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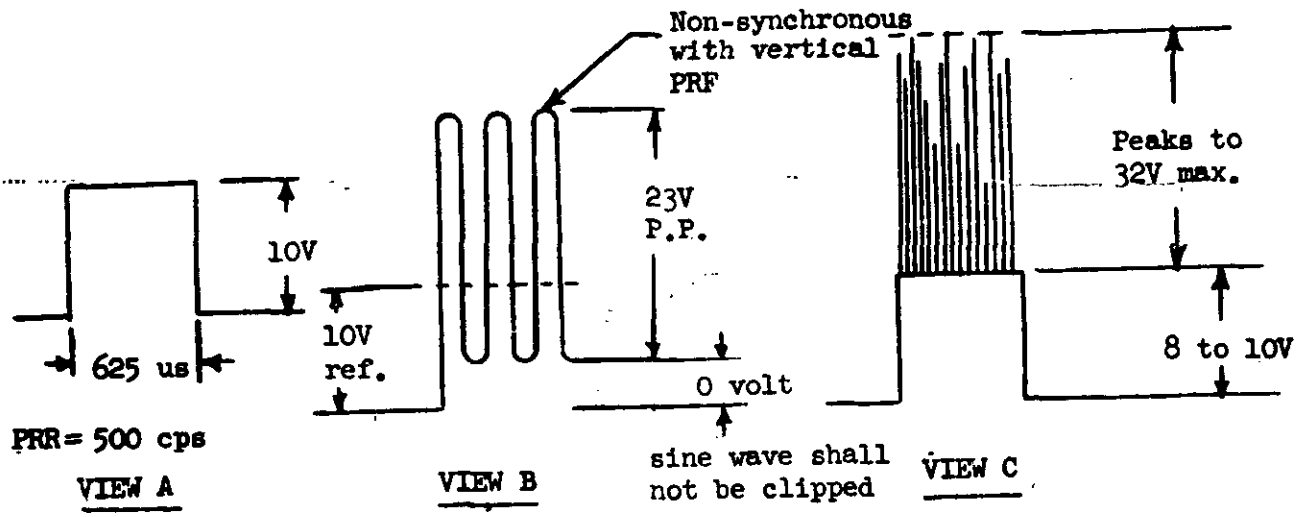


Figure 6 - Grid drive test waveforms.

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